

HSC Mathematics Advanced (Year 12)

Official NESA Standard Solutions & Marking Guidelines

Paper: MAD 12 Revision
Assessment: Trial Exam Preparation
Syllabus: NSW NESA Stage 6
Standard: NESA Criteria Solutions
Formulas: NESA Reference Sheet

QUESTION 1 | Integral Calculus — Linear Substitution

[2 Marks]

Question: Find $\int \sqrt{4x + 1} \, dx$.

NESA Reference Sheet / Key Rule:

$$\int (ax + b)^n \, dx = [(ax + b)^{n+1}] / [a(n + 1)] + c, \text{ where } n \neq -1$$

Detailed Working (NESA Standard Solution):

- Rewrite in index form: $\int \sqrt{4x + 1} \, dx = \int (4x + 1)^{1/2} \, dx$
- Identify parameters: $a = 4$, $b = 1$, $n = 1/2$.
- Apply primitive rule:
 $\int (4x + 1)^{1/2} \, dx = [(4x + 1)^{3/2}] / [4 \times (3/2)] + C = 1/6 (4x + 1)^{3/2} + C$

Final Answer: $1/6 (4x + 1)^{3/2} + C$ (or $[\sqrt{(4x + 1)^3}] / 6 + C$)

Marks	Marking Criteria Breakdown
1 Mark	Expresses integrand in index form and correctly applies linear rule (divides by 4 and adds 1 to power).
1 Mark	Simplifies coefficient to 1/6 and includes constant of integration + C.

QUESTION 2 | Differential Calculus — Increasing Functions

[2 Marks]

Question: For the function $f(x) = x^3 - 9x$, state the domain where $f(x)$ is increasing.

NESA Reference Sheet / Key Rule:

A function $f(x)$ is increasing where $f'(x) \geq 0$ (or strictly increasing where $f'(x) > 0$).

Detailed Working (NESA Standard Solution):

- Differentiate $f(x)$: $f'(x) = 3x^2 - 9$
- Condition for increasing: $f'(x) \geq 0 \Rightarrow 3(x^2 - 3) \geq 0 \Rightarrow x^2 \geq 3$
- Solve quadratic inequality: $x \leq -\sqrt{3}$ or $x \geq \sqrt{3}$ $((-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty))$.
(Strict inequalities $x < -\sqrt{3}$ or $x > \sqrt{3}$ are also accepted by NESA).

Final Answer: $x \leq -\sqrt{3}$ or $x \geq \sqrt{3}$ (or $(-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty)$)

Marks	Marking Criteria Breakdown
1 Mark	Differentiates correctly to obtain $f'(x) = 3x^2 - 9$.
1 Mark	Solves $f'(x) \geq 0$ to state the correct domain in terms of $\pm\sqrt{3}$.

QUESTION 3 | Integral Calculus — Logarithmic Forms

[2 Marks]

Question: Find $\int [e^{3x} / (e^{3x} + 1)] \, dx$.

NESA Reference Sheet / Key Rule:

$$\int [f'(x) / f(x)] \, dx = \ln |f(x)| + c$$

Detailed Working (NESA Standard Solution):

- Identify derivative of denominator: Let $f(x) = e^{3x} + 1 \Rightarrow f'(x) = 3e^{3x}$.
- Balance constant factor: $\int [e^{3x} / (e^{3x} + 1)] \, dx = 1/3 \int [3e^{3x} / (e^{3x} + 1)] \, dx$
- Evaluate integral: $= 1/3 \ln(e^{3x} + 1) + C$ (Absolute values not needed as $e^{3x} + 1 > 0$).

Final Answer: $1/3 \ln(e^{3x} + 1) + C$

Marks	Marking Criteria Breakdown
1 Mark	Recognises $f'(x)/f(x)$ form and balances with multiplier 1/3.
1 Mark	Obtains correct primitive with + C.

QUESTION 4 | Calculus — Chain Rule & Reverse Chain Rule**[4 Marks]**Question: a. Find $d/dx (\sqrt{x} + 1)^3$ b. Hence, find $\int [(\sqrt{x} + 1)^2 / (2\sqrt{x})] dx$

NESA Reference Sheet / Key Rule:

Chain Rule: $d/dx [g(x)]^n = n[g(x)]^{n-1} g'(x)$ **Detailed Working (NESA Standard Solution):**

Part a:

- Inner function: $g(x) = \sqrt{x} + 1 = x^{1/2} + 1 \Rightarrow g'(x) = 1/(2\sqrt{x})$.
- $d/dx [(\sqrt{x} + 1)^3] = 3(\sqrt{x} + 1)^2 \times [1/(2\sqrt{x})] = [3(\sqrt{x} + 1)^2] / (2\sqrt{x})$

Part b (Using 'Hence'):

- From Part a: $d/dx [(\sqrt{x} + 1)^3] = 3 \times [(\sqrt{x} + 1)^2 / (2\sqrt{x})]$.
- Integrate: $3 \int [(\sqrt{x} + 1)^2 / (2\sqrt{x})] dx = (\sqrt{x} + 1)^3 + C_1$
- $\int [(\sqrt{x} + 1)^2 / (2\sqrt{x})] dx = 1/3 (\sqrt{x} + 1)^3 + C$

Final Answer: a. $[3(\sqrt{x} + 1)^2] / (2\sqrt{x})$ b. $1/3 (\sqrt{x} + 1)^3 + C$

Marks	Marking Criteria Breakdown
2 Marks (a)	1 mark for chain rule application; 1 mark for correct simplified derivative.
2 Marks (b)	1 mark for relating integral to derivative of Part a; 1 mark for primitive with + C.

QUESTION 5 | Sequences and Series — Arithmetic Progression (AP)**[3 Marks]**

Question: On day 1 of harvest, orchard produces 1630 kg fruit. Next day: 1605 kg, decreasing by same amount each day. On what day does production first fall below 600 kg?

NESA Reference Sheet / Key Rule:

n-th term of an AP: $T_n = a + (n - 1)d$ **Detailed Working (NESA Standard Solution):**

1. Identify AP parameters: $a = 1630$, $d = 1605 - 1630 = -25$.
2. Express general term: $T_n = 1630 + (n - 1)(-25) = 1655 - 25n$.
3. Solve inequality $T_n < 600$:
 $1655 - 25n < 600 \Rightarrow -25n < -1055 \Rightarrow n > (-1055)/(-25) = 42.2$
4. Conclusion: Since n must be an integer, $n = 43$.
 Check: $T_{42} = 1630 + 41(-25) = 605$ kg; $T_{43} = 1630 + 42(-25) = 580$ kg < 600 kg.

Final Answer: Day 43 (or the 43rd day)

Marks	Marking Criteria Breakdown
1 Mark	Identifies $a = 1630$, $d = -25$ and states formula for T_n .
1 Mark	Sets up $T_n < 600$ and solves inequality to find $n > 42.2$.
1 Mark	Correctly concludes Day 43.

QUESTION 6 | Differential Calculus — Curve Sketching & Nature of Points**[7 Marks]**

Question: Consider the curve $y = x^3 - 6x^2 + 9x + 3$.

(a) Find location and nature of any stationary points.

(b) Find the point of inflection.

(c) Sketch the curve labelling stationary points, inflection and y-intercept (x-intercepts not required).

NESA Reference Sheet / Key Rule:

Stationary points: $y' = 0$. Nature: $y'' < 0$ (Local Max), $y'' > 0$ (Local Min). Inflection: $y'' = 0$ with concavity change.

Detailed Working (NESA Standard Solution):

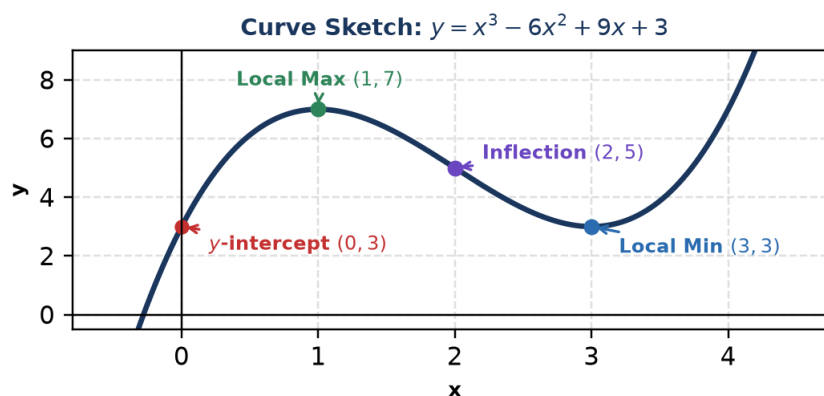
(a) Stationary Points:

- $y' = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0 \Rightarrow x = 1$ or $x = 3$.
- At $x = 1$: $y = 1 - 6 + 9 + 3 = 7 \Rightarrow$ Point $(1, 7)$.
- At $x = 3$: $y = 27 - 54 + 27 + 3 = 3 \Rightarrow$ Point $(3, 3)$.
- Second derivative: $y'' = 6x - 12$.
- At $x = 1$: $y'' = 6(1) - 12 = -6 < 0 \Rightarrow$ Local Maximum at $(1, 7)$.
- At $x = 3$: $y'' = 6(3) - 12 = +6 > 0 \Rightarrow$ Local Minimum at $(3, 3)$.

(b) Point of Inflection:

- $y'' = 6x - 12 = 0 \Rightarrow x = 2$.
- At $x = 2$: $y = 8 - 24 + 18 + 3 = 5 \Rightarrow$ Point $(2, 5)$.
- Testing concavity: For $x = 1$, $y'' = -6 < 0$ (concave down); for $x = 3$, $y'' = +6 > 0$ (concave up).
- Since concavity changes across $x = 2$, $(2, 5)$ is a point of inflection.

(c) Curve Sketch: y-intercept: at $x = 0$, $y = 3 \Rightarrow$ Point $(0, 3)$.



Final Answer: (a) Local Max at $(1, 7)$; Local Min at $(3, 3)$

(b) Point of Inflection at $(2, 5)$

(c) See plot below with $(0, 3)$, $(1, 7)$, $(2, 5)$, $(3, 3)$ clearly labelled.

Marks	Marking Criteria Breakdown
3 Marks (a)	1 mark for derivative & stationary x-values; 1 mark for coordinates $(1, 7)$ & $(3, 3)$; 1 mark for nature test.
2 Marks (b)	1 mark for finding $x = 2$, $y = 5$; 1 mark for verifying change of concavity across $x = 2$.
2 Marks (c)	1 mark for correct cubic shape; 1 mark for plotting and labelling $(0, 3)$, $(1, 7)$, $(2, 5)$, $(3, 3)$.

QUESTION 7 | Calculus — Tangents & Normals to Logarithmic Curves**[3 Marks]**

Question: A curve has equation $y = \log_e(x^3 - 5)$. Find equation of normal to curve at point $x = 2$.

NESA Reference Sheet / Key Rule:

$$\frac{d}{dx} [\ln f(x)] = f'(x) / f(x) \quad | \quad m_{\text{normal}} = -1 / m_{\text{tangent}} \quad | \quad y - y_1 = m(x - x_1)$$

Detailed Working (NESA Standard Solution):

- Point: At $x = 2$, $y = \ln(2^3 - 5) = \ln 3 \Rightarrow$ Point on curve is $(2, \ln 3)$.
- Gradient: $dy/dx = 3x^2/(x^3 - 5) \Rightarrow m_{\text{tangent}} = [3(2)^2]/(2^3 - 5) = 12/3 = 4$.
- Normal gradient: $m_{\text{normal}} = -1/4$.
- Equation: $y - \ln 3 = -1/4(x - 2) \Rightarrow x + 4y - (2 + 4 \ln 3) = 0$ (or $y = -1/4 x + 1/2 + \ln 3$)

Final Answer: $x + 4y - (2 + 4 \ln 3) = 0$ (or $y = -1/4 x + 1/2 + \ln 3$)

Marks	Marking Criteria Breakdown
1 Mark	Finds $y = \ln 3$ and differentiates correctly to $dy/dx = 3x^2/(x^3 - 5)$.
1 Mark	Calculates $m_{\text{tangent}} = 4$ and $m_{\text{normal}} = -1/4$.
1 Mark	Forms correct exact equation of the normal line.

QUESTION 8 | Trigonometric Functions — Quadratic Equations**[3 Marks]**

Question: Solve $2\cos^2 x + 3\cos x + 1 = 0$ for $0 \leq x \leq 2\pi$.

NESA Reference Sheet / Key Rule:

Factorise quadratic in $\cos x$; solve in radians across quadrants $[0, 2\pi]$.

Detailed Working (NESA Standard Solution):

- Factorise quadratic: $(2\cos x + 1)(\cos x + 1) = 0 \Rightarrow \cos x = -1/2$ or $\cos x = -1$
- Radian solutions:
 - $\cos x = -1 \Rightarrow x = \pi$
 - $\cos x = -1/2$ (ref angle $\pi/3$, Q II & III): $x = \pi - \pi/3 = 2\pi/3$; $x = \pi + \pi/3 = 4\pi/3$

Final Answer: $x = 2\pi/3, \pi, 4\pi/3$

Marks	Marking Criteria Breakdown
1 Mark	Factorises quadratic correctly into $(2\cos x + 1)(\cos x + 1) = 0$.
1 Mark	Obtains $x = \pi$ from $\cos x = -1$.
1 Mark	Obtains both correct quadrant solutions $x = 2\pi/3$ and $x = 4\pi/3$.

QUESTION 9 | Functions — Absolute Value Equations**[2 Marks]**Question: Solve the equation $|2x + 3| = 7$.

NESA Reference Sheet / Key Rule:

$$|X| = a \iff X = a \text{ or } X = -a \text{ (for } a \geq 0 \text{)}$$

Detailed Working (NESA Standard Solution):

1. Split into two linear cases: $2x + 3 = 7$ or $2x + 3 = -7$
2. Solve: $2x = 4 \Rightarrow x = 2$; $2x = -10 \Rightarrow x = -5$

Final Answer: $x = 2$ or $x = -5$ **Marks Marking Criteria Breakdown**1 Mark Sets up two linear equations $2x + 3 = \pm 7$.1 Mark Obtains both correct solutions $x = 2$ and $x = -5$.**QUESTION 10 | Calculus — Product and Chain Rules****[3 Marks]**Question: Find derivative of $y = \sin 5x \tan(e^x)$.

NESA Reference Sheet / Key Rule:

$$(uv)' = u'v + uv' \quad | \quad d/dx[\sin kx] = k \cos kx \quad | \quad d/dx[\tan f(x)] = f'(x) \sec^2 f(x)$$

Detailed Working (NESA Standard Solution):

1. Differentiate components: $u = \sin 5x \Rightarrow u' = 5 \cos 5x$; $v = \tan(e^x) \Rightarrow v' = e^x \sec^2(e^x)$.
2. Product rule: $dy/dx = 5 \cos 5x \tan(e^x) + e^x \sin 5x \sec^2(e^x)$

Final Answer: $dy/dx = 5 \cos 5x \tan(e^x) + e^x \sin 5x \sec^2(e^x)$ **Marks Marking Criteria Breakdown**

1 Mark Applies product rule formula correctly.

1 Mark Differentiates either u or v correctly via chain rule.

1 Mark Obtains correct complete derivative expression.

QUESTION 11 | Functions — Composite Functions (Domain and Range)**[3 Marks]**Question: Given $f(x) = \sqrt{x + 1}$ and $g(x) = 2x - 3$. Find domain and range of $f(g(x))$.

NESA Reference Sheet / Key Rule:

Radical of square root must be non-negative (≥ 0); principal square root outputs non-negative values.**Detailed Working (NESA Standard Solution):**

1. Form composite function: $f(g(x)) = f(2x - 3) = \sqrt{(2x - 3) + 1} = \sqrt{2x - 2}$
2. Determine Domain: $2x - 2 \geq 0 \Rightarrow 2x \geq 2 \Rightarrow x \geq 1$ (or $[1, \infty)$).
3. Determine Range: At $x = 1$, $f(g(1)) = 0$. As $x \rightarrow \infty$, $\sqrt{2x - 2} \rightarrow \infty$. Since $\sqrt{\cdot}$ is principal square root: $y \geq 0$ (or $[0, \infty)$).

Final Answer: Domain: $x \geq 1$ ($[1, \infty)$) | Range: $y \geq 0$ ($[0, \infty)$)**Marks Marking Criteria Breakdown**1 Mark Forms composite expression $f(g(x)) = \sqrt{2x - 2}$.1 Mark Finds correct domain $x \geq 1$.1 Mark Finds correct range $y \geq 0$.

QUESTION 12 | Statistical Analysis — Linear Regression and Extrapolation**[6 Marks]**

Question: Tom surveyed age (x) and weekly wage (W):

Age (x): 18, 45, 28, 15, 32, 68 | Wage (W): 715, 2350, 1530, 438, 1690, 1320

- Find correlation coefficient r and explain strength.
- Find least-squares regression line in form $W = Bx + A$ (A, B integers).
- Could your equation be used to make valid estimate for ages > 68 or < 15 ? Justify.

NESA Reference Sheet / Key Rule:

Pearson's r measures linear correlation; Least-squares regression line: $W = Bx + A$

Detailed Working (NESA Standard Solution):

Part i (Correlation coefficient):

- Calculator stats: $r \approx 0.5263$ (or 0.53 to 2 d.p.).
- Strength: There is a moderate positive linear correlation between age and weekly wage.

Part ii (Regression line):

- Slope: $B = 18.4795 \Rightarrow B = 18$ (nearest integer).
- Intercept: $A = 706.0378 \Rightarrow A = 706$ (nearest integer).
- Regression equation: $W = 18x + 706$.

Part iii (Validity / Extrapolation):

- No, it cannot be used to make valid estimates.
- Justification: Estimating wages for ages outside $[15, 68]$ is extrapolation. Extrapolation is unreliable because linear trend cannot be assumed to hold beyond the sample. In reality, people older than 68 typically retire and wage income drops substantially, while individuals younger than 15 are children who cannot legally work full-time yet model predicts \$706/week at age 0.

Final Answer: i. $r \approx 0.53$ (moderate positive linear correlation)

ii. $W = 18x + 706$

iii. No; extrapolation outside $[15, 68]$ is unreliable and unrealistic in context.

Marks	Marking Criteria Breakdown
2 Marks (i)	1 mark for $r \approx 0.53$; 1 mark for describing as moderate positive correlation.
2 Marks (ii)	1 mark for finding slope & intercept; 1 mark for integer equation $W = 18x + 706$.
2 Marks (iii)	1 mark for identifying extrapolation outside $[15, 68]$; 1 mark for valid practical justification.

QUESTION 13 | Calculus / Differential Equations — Exponential Growth**[4 Marks]**

Question: $dP/dt = 1200 e^{0.3t}$. Population is 5000 after 1 month. Determine how long it will take to reach viable stage of at least 100 000 (nearest month).

NESA Reference Sheet / Key Rule:

$$\int e^{kt} dt = (1/k) e^{kt} + c$$

Detailed Working (NESA Standard Solution):

1. Integrate rate equation: $P(t) = \int 1200 e^{0.3t} dt = (1200 / 0.3) e^{0.3t} + C = 4000 e^{0.3t} + C$
2. Evaluate C ($t = 1$, $P = 5000$): $5000 = 4000 e^{0.3} + C \Rightarrow C = 5000 - 4000 e^{0.3} \approx -399.435$
3. Population model: $P(t) = 4000 e^{0.3t} - 399.435$
4. Solve for $P(t) = 100\,000$:
 $4000 e^{0.3t} = 100\,399.435 \Rightarrow e^{0.3t} = 25.09986 \Rightarrow 0.3t = \ln(25.09986) \approx 3.22286$
 $t = 3.22286 / 0.3 \approx 10.74$ months $\Rightarrow$ 11 months (nearest month).

Final Answer: 11 months

Marks	Marking Criteria Breakdown
1 Mark	Integrates correctly to $P(t) = 4000 e^{0.3t} + C$.
1 Mark	Substitutes $t = 1$, $P = 5000$ to find $C \approx -399.44$.
1 Mark	Sets $P(t) = 100\,000$ and solves for $t \approx 10.74$.
1 Mark	Rounds to nearest month (11 months).

QUESTION 14 | Continuous Probability — Normal Distribution**[3 Marks]**

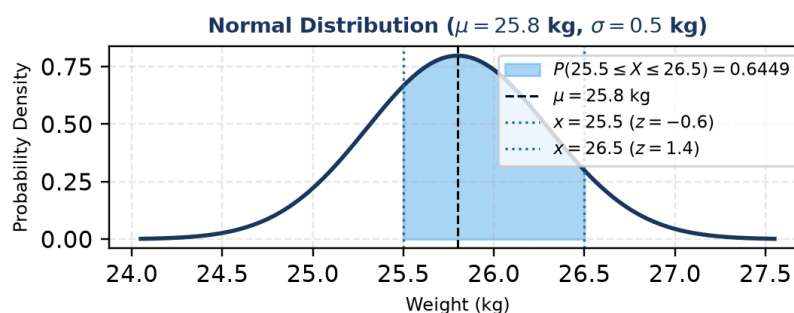
Question: Weights of bags of gravel have normal distribution with mean 25.8 kg and standard deviation 0.5 kg. Using the provided z-table, find the probability that a bag weighs between 25.5 kg and 26.5 kg.

NESA Reference Sheet / Key Rule:

$$z = (x - \mu) / \sigma \quad | \quad P(a \leq X \leq b) = \Phi(z_2) - \Phi(z_1) \quad | \quad \Phi(-z) = 1 - \Phi(z)$$

Detailed Working (NESA Standard Solution):

1. Convert boundaries to z-scores: ($\mu = 25.8$, $\sigma = 0.5$)
 - For $x_1 = 25.5$ kg: $z_1 = (25.5 - 25.8)/0.5 = -0.6$
 - For $x_2 = 26.5$ kg: $z_2 = (26.5 - 25.8)/0.5 = 1.4$
2. Express via standard normal CDF $\Phi(z)$: $P(25.5 \leq X \leq 26.5) = \Phi(1.4) - \Phi(-0.6)$
3. Read values from table:
 - Row 1., Col .4 $\Rightarrow \Phi(1.4) = 0.9192$.
 - By symmetry: $\Phi(-0.6) = 1 - \Phi(0.6)$. Row 0., Col .6 $\Rightarrow \Phi(0.6) = 0.7257 \Rightarrow \Phi(-0.6) = 1 - 0.7257 = 0.2743$.
4. Calculate probability: $P(-0.6 \leq Z \leq 1.4) = 0.9192 - 0.2743 = 0.6449$ (or 64.49%).

**Final Answer: 0.6449 (or 64.49%)**

Marks	Marking Criteria Breakdown
1 Mark	Standardises both values to $z_1 = -0.6$ and $z_2 = 1.4$.
1 Mark	Correctly extracts $\Phi(1.4) = 0.9192$ and computes $\Phi(-0.6) = 0.2743$ from table.
1 Mark	Obtains correct difference 0.6449.

QUESTION 15 | Calculus / Optimization — Mensuration & Cost Minimization**[6 Marks]**

Question: Open-ended triangular prism tent. Rods cost \$9.50/m. Top and bottom edges: length x . Sloping edges: length y . Covered by one rectangular sheet of fabric of area 24 m^2 .

(a) Show that the total length l of rods is $l = 3x + 48/x$.

(b) Find x for which l is a minimum.

(c) What is the minimum cost of the frame?

NESA Reference Sheet / Key Rule:

Area of fabric sheet = $2xy = 24$. Optimization: $dl/dx = 0$; verify $d^2l/dx^2 > 0$.

Detailed Working (NESA Standard Solution):

(a) Total length formula:

- Rods of length x : 1 top ridge rod + 2 bottom ground rods = 3 rods of length x .
- Rods of length y : 2 sloping rods at front end + 2 at back end = 4 rods of length y .
- Total length: $l = 3x + 4y$.
- Fabric sheet covers both sloping roofs: Area = $x(2y) = 2xy = 24 \Rightarrow y = 24/(2x) = 12/x$.
- Substitute y into l : $l = 3x + 4(12/x) = 3x + 48/x$ (Shown).

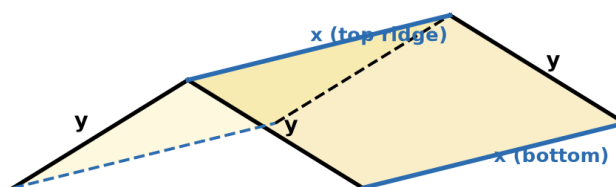
(b) Minimum length:

- $dl/dx = 3 - 48/x^2 = 0 \Rightarrow 3x^2 = 48 \Rightarrow x^2 = 16 \Rightarrow x = 4 \text{ m}$ (since $x > 0$).
- Nature verification: $d^2l/dx^2 = 96/x^3$. At $x = 4$: $d^2l/dx^2 = 96/64 = 1.5 > 0 \Rightarrow$ Minimum at $x = 4 \text{ m}$.

(c) Minimum Cost:

- Total minimum length: $l(4) = 3(4) + 48/4 = 12 + 12 = 24 \text{ m}$.
- Frame Cost = $24 \text{ m} \times \$9.50/\text{m} = \228.00 .

Open-Ended Triangular Prism Tent Frame



Final Answer: (a) Shown: $l = 3x + 48/x$

(b) $x = 4 \text{ m}$ (confirmed minimum via second derivative test)

(c) \$228.00

Marks	Marking Criteria Breakdown
2 Marks (a)	1 mark for $l = 3x + 4y$ and $2xy = 24$; 1 mark for substitution to obtain $l = 3x + 48/x$.
3 Marks (b)	1 mark for derivative $dl/dx = 3 - 48/x^2$; 1 mark for $x = 4$; 1 mark for verifying minimum.
1 Mark (c)	1 mark for finding $l = 24 \text{ m}$ and multiplying by $\$9.50$ to get $\$228.00$.

QUESTION 16 | Financial Mathematics — Future Value of Annuities**[3 Marks]**

Question: Annie wants to save \$10 000 over 2 years by investing equal payments every 6 months at 8% p.a. compounded half-yearly. Using the provided future value interest table, what should be the value of each payment?

NESA Reference Sheet / Key Rule:

Future Value = Payment (M) $\times$ Table Factor | n = compounding periods, r = rate per period

Detailed Working (NESA Standard Solution):

1. Compounding periods (n) and rate (r):

- Time = 2 years, compounding half-yearly (every 6 months): $n = 2 \times 2 = 4$ periods.
- Interest rate per half-year: $r = (8\% \text{ p.a.}) / 2 = 4\% \text{ per period}$.

2. Extract Future Value Interest Factor from table:

- Row: Period = 4, Column: 4% $\Rightarrow$ Factor = 4.2465.

3. Calculate payment (M):

- $\$10\,000 = M \times 4.2465 \Rightarrow M = 10\,000 / 4.2465 \approx 2354.880489... \Rightarrow \2354.88 (nearest cent).

Final Answer: \$2354.88

Marks	Marking Criteria Breakdown
1 Mark	Correctly identifies $n = 4$ periods and $r = 4\%$ per period.
1 Mark	Extracts correct factor 4.2465 from table.
1 Mark	Calculates $M = 10\,000 / 4.2465 = \$2354.88$.

Summary of All Final Answers (Questions 1 to 16)

Quick reference table for checking and cross-referencing against NESA marking benchmarks.

Q#	Topic / Syllabus Area	Final Answer (NESA Standard)
1	Calculus — Linear Substitution	$\frac{1}{6} (4x + 1)^{3/2} + C$
2	Calculus — Increasing Functions	$x \leq -\sqrt{3}$ or $x \geq \sqrt{3}$ $((-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty))$
3	Calculus — Logarithmic Integration	$\frac{1}{3} \ln(e^{3x} + 1) + C$
4a	Calculus — Chain Rule	$[3(\sqrt{x} + 1)^2] / (2\sqrt{x})$
4b	Calculus — Reverse Chain Rule	$\frac{1}{3} (\sqrt{x} + 1)^3 + C$
5	Sequences & Series — AP	Day 43
6a	Calculus — Stationary Points	Local Max: (1, 7); Local Min: (3, 3)
6b	Calculus — Point of Inflection	Point of Inflection: (2, 5)
6c	Calculus — Curve Sketching	Cubic plotted through (0, 3), (1, 7), (2, 5), (3, 3)
7	Calculus — Tangents & Normals	$x + 4y - (2 + 4 \ln 3) = 0$ (or $y = -\frac{1}{4}x + \frac{1}{2} + \ln 3$)
8	Trigonometry — Quadratic Eqn	$x = 2\pi/3, \pi, 4\pi/3$
9	Functions — Absolute Value	$x = 2$ or $x = -5$
10	Calculus — Product & Chain Rule	$5 \cos 5x \tan(e^x) + e^x \sin 5x \sec^2(e^x)$
11	Functions — Domain & Range	Domain: $x \geq 1$ $([1, \infty))$; Range: $y \geq 0$ $([0, \infty))$
12i	Statistics — Correlation	$r \approx 0.53$ (moderate positive linear correlation)
12ii	Statistics — Regression Line	$W = 18x + 706$
12iii	Statistics — Validity & Extrapolation	No; extrapolation outside [15, 68] is unreliable
13	Differential Eqn — Exponential	11 months
14	Continuous Probability — Normal Dist	0.6449 (or 64.49%)
15a	Optimization — Total Length	Shown: $l = 3x + 48/x$
15b	Optimization — Minimum Length	$x = 4$ m
15c	Optimization — Minimum Cost	\$228.00
16	Financial Maths — Annuities	\$2354.88